

ADAPTIVE CONFORMAL SETS UNDER DISTRIBUTION SHIFT: USING ENSEMBLE DISAGREEMENT AS AN EPISTEMIC NORMALIZER FOR EEL-GRASS SEGMENTATION UNDER TEMPORAL DRIFT

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TITLE: Adaptive Conformal Sets Under Distribution Shift: Using Ensemble Disagreement as an Epistemic Normalizer for Eelgrass Segmentation under Temporal Drift

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ABSTRACT

Adaptive Conformal Sets Under Distribution Shift: Using Ensemble Disagreement as an Epistemic Normalizer for Eelgrass Segmentation under Temporal Drift

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Semantic segmentation of eelgrass (*Zostera marina*) from high-resolution drone-based imagery is essential for coastal habitat monitoring, restoration, and management, as these habitats see rapid changes due to climate change and human-induced influences. However, the reliability of generalizing these classification models not only relies on high-accuracy segmentation but also on rigorous uncertainty quantification that holds up when conditions change across years or locations. Conformal prediction (CP) converts a classifier’s output into prediction sets with finite-sample marginal coverage; however, the standard score $s = 1 - p_y$ can under-cover in hard or out-of-distribution (OOD) regions under temporal drift. Inspired by heteroskedastic conformal regression, this study proposes normalizing conformal scores by an epistemic proxy, ensemble disagreement. These scores re-scale nonconformity by a data-driven estimation of local difficulty learned on the calibration set, so that the global quantile is conservative where OOD risk is high. This study evaluates (i) a parametric form $s' = (1 - p_y)/(1 + \lambda V)$, where V is ensemble variance of probabilities, and (ii) a nonparametric normalization $s' = s/\hat{a}(V)$, where $\hat{a}(V) = E[s|V]$ is estimated through quantile binning and interpolation. On drone imagery of the Morro Bay estuary, California (2018-2022), the models were trained on 2018-2021, calibrated on 2021, and evaluated on 2022 on OOD points. Normalized scores recover much of the lost coverage seen by vanilla split conformal prediction and approximate target, increased the percentage of singletons, and reduced the spatial coverage variability. The study provides a simple-to-implement drop-in framework to regain coverage in difficult regions and out-of-distribution data without retraining or labels at test time.

Keywords: conformal prediction, adaptive sets, uncertainty, eelgrass, remote sensing.

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Chapter 1

INTRODUCTION

Eelgrass (*Zostera marina*) is a seagrass species of temperate waters that provides an anchor to coastal ecosystems by providing sediment stabilization, carbon sequestration, eliminating contamination, and providing habitats for protected species. Warming oceans, storm regimes, and local disturbance have quickly reshaped the intertidal morphology and habitat extent. As these threats increase, there is a need to monitor changes to determine appropriate environmental management. In recent years, deep networks have made this intertidal mapping practical in complex imagery, achieving strong pixel-wise accuracy in areas such as Morro Bay, California (Tallam et al., 2023). However, the reliability of general utilization of these classification models not only relies on high-accuracy segmentation, but also on rigorous uncertainty quantification under temporal and spatial distribution shift. Imagery collected over different years, at different tides or seasons, can erode the model’s calibration, causing it to be over-confident. This study builds off these models to create a post hoc uncertainty framework.

Uncertainty quantification is a requirement for flagging ambiguous areas. To answer this, split conformal prediction provides an appealing framework. It offers a wrapping of any probabilistic predictor to produce prediction sets with finite-sample marginal coverage under exchangeability, simply by using a calibration set and a quantile nonconformity score (Angelopoulos & Bates, 2022). In classification, a natural choice of nonconformity score is $s = 1 - p_y$, yielding a prediction set of all labels whose scores do not exceed a global threshold learned on calibration. With stable data, this can be simple and effective; however, in reality, the distribution of scores can differ by difficulty. A threshold calibrated over one year can underperform when a distributional shift occurs, such as temporal drift.

To remedy this, it is necessary to make the conformal prediction adaptive to drift or local difficulty. First, work in adaptive or locally weighted CP in regression has shown that normalizing scores by a measure of local difficulty can make a single global quantile more appropriate across heterogeneous inputs, which improves approximate conditional behavior (Chernozhukov, Wüthrich, & Zhu, 2021; Dewolf, De Baets, & Waegeman, 2025; Lei, G’Sell, Rinaldo, Tibshirani, & Wasserman, 2018). Second, research on uncertainty in deep learning has demonstrated that diversified deep ensembles provide practical, shift-aware signals of difficulty that tend to grow on out-of-distribution inputs (Lakshminarayanan, Pritzel, & Blundell, 2017). Using these ideas, this study implements a simple strategy for rescaling the classification score by a calibration-learned difficulty proxy, so that the global threshold becomes conservative under drift, without the need for recalibration or labels.

This thesis develops this design for eelgrass segmentation of multi-year drone imagery. The proposed normalizers are deliberately simple to adopt into existing pipelines. This includes: 1. a linear normalization $s' = (1 - p_y)/(1 + \lambda V)$ where V summarizes ensemble disagreement and λ controls degree of conservativeness and 2. a learned normalizer $s' = s/\hat{a}(V)$ where $\hat{a}(V) = E[s|V]$ is estimated on the calibration split via quantile binning and interpolation. Both approaches retain standard CP workflow and require no labels at test time or retraining.

Chapter 2

BACKGROUND AND MOTIVATION

Effective eelgrass monitoring requires models that both provide accurate predictions, as well as how certain they are for those predictions. This section reviews the uncertainty concepts used in deep learning.

2.1 Uncertainty in Deep Learning

Uncertainty in predictive modeling is commonly broken up into two components: *epistemic* and *aleatoric*. These two components collectively comprise the total uncertainty in a model’s predictions. Aleatoric uncertainty represents the noise inherent in observations, which is often impossible to remove, even with the addition of more data (e.g. sensor noise or true label ambiguity). Even a perfect model cannot remove it. Epistemic uncertainty represents the uncertainty of the model itself, which can be reduced with additional or higher-quality data (Kendall & Gal, 2017).

For modern vision models, raw softmax probabilities are often miscalibrated, meaning the outputted confidence does not match empirical accuracy. This can be especially prevalent under shift, where models can be confident yet wrong. Post-hoc calibration, such as temperature scaling, rescales logits via $\text{softmax}(z/T)$ and improves in-distribution calibration without changing accuracy (Guo, Pleiss, Sun, & Weinberger, 2017). Under shift, however, calibration is not enough, as the structure of the errors can change. Thus, it requires an uncertainty representation that integrates decision rules that signal the difficulty of the prediction.

2.2 Deep Ensembles

Deep ensembles can address uncertainty by aggregating predictions from models trained independently with different initializations or architectures. For estimating epistemic uncertainty, member disagreements can be calculated by considering the variance in member probabilities or logits. Given members $\{f_k\}_{k=1}^K$ that produce $p^{(k)}(x)$, $V(x) = \text{Var}_k(p_y^{(k)}(x))$. Ensembles have not only been found to be more robust under shift, but taking epistemic uncertainty specifically into account can greatly increase a model’s predictive uncertainty under dataset shift (Ovadia et al., 2019). Additionally, ensembles require no specialized inference and can easily integrate with existing training code, providing a difficulty proxy for uncertainty. This does come at the additional computational cost of training additional models. However, many models are often trained in pursuit of classification accuracy, which, when ensembled, can outperform individual models.

2.3 Semantic Segmentation in Environmental Monitoring

Semantic image segmentation is a deep learning neural network technique that assigns class labels to each pixel within an image, delineating objects. Recent work demonstrates that deep networks can accurately classify intertidal eelgrass from drone imagery at useful resolutions (Tallam et al., 2023). Environmental monitoring can pose unique challenges, as not only are annotations costly and sometimes ambiguous, but the habitat is constantly shifting over seasonal and annual time scales. Even at the same site, small variations such as time of day, tidal height, etc, can alter the conditions of the imagery.

2.4 Conformal Prediction Foundations

Conformal prediction provides a model and distribution-free technique to convert probability scores into prediction sets with finite-sample marginal coverage under exchangeability (Angelopoulos & Bates, 2022). For a classifier with probabilities $p(x)$, a simple *nonconformity*

score is $s = 1 - p_y$, which is small when the model is confident in the true class. In split conformal prediction, scores are computed on a calibration set C of size n and the finite-sample-corrected quantile is selected $\hat{q}_\alpha = \text{Quantile}_{[(n+1)(1-\alpha)]/n}(\{s_i\})$, then the set value prediction is outputed

$$C_\alpha(x) = \{y : s \leq \hat{q}_\alpha\}$$

This procedure guarantees $P\{Y \in C_\alpha(X)\} \geq 1 - \alpha$ for new points exchangeable with C , regardless of the underlying model. Two limitations of conformal prediction motivate this thesis. The first limitation is that the guarantee is marginal. This means there is no guarantee of conditional coverage, that the per class coverage will be $\geq$ target, and that coverage is not guaranteed across differing inputs, such as varying difficulty. Second, conformal prediction has mathematical validity due to the concept of symmetry (data could have been seen in any order) as long as the data is exchangeable. Coverage guarantees are lost as distribution drifts, and the model can become overly confident, leading to under-coverage. To address these problems, we need to use an adaptive variant of conformal prediction, which is more conservative for harder inputs.

2.5 Adaptive or Shift-Aware CP

There is a substantial amount of literature that aims to make CP adaptive to heterogeneity or robust during dataset drift.

In regression, locally-weighted CP normalizes residuals by an estimate of local spread to equalize scores for a global quantile under heterogeneity (Lei et al., 2018). Distributional conformal prediction (DCP) transforms the score using an estimate of the conditional cumulative distribution function. If the distribution of the score is stable conditional on covariates, then a single global quantile should be appropriate across those covariates (Chernozhukov et al., 2021).

The two ideas loosely connect to this variance score normalization. Let $S = s(X, Y)$ denote the standard nonconformity score and let $V = V(X)$ be a one-dimensional proxy from the deep ensemble (member-probability variance after temperature scaling). We can use this proxy to stabilize the score with respect to V , rather than estimating an ultra-high-dimensional $F_{S|X}$ as in DCP. Additionally, we do not estimate a full conditional CDF transform and instead normalize under a multiplicative difficulty assumption

$$S = a(V)U, \quad a(V) > 0, \quad U \perp\!\!\!\perp V.$$

Then, $S' = S/a(V) \approx U$, so $S'|V$ becomes approximately invariant, and locally tuned to difficulty. Normalization targets the actual failure under shift, that the global threshold will become too lenient for the hard OOD inputs. On the calibration set, we estimate $\hat{a} \approx E[S|V = v]$ via quantile binning and smooth interpolation, and then conformalize the normalized score $S' = S/\hat{a}(V)$. Thus, if the conditional distribution $S|V = v$ is roughly the same across years:

$$S | V = v(\text{calibration}) \approx S | V = v(\text{test}),$$

then we expect the normalized scores to be approximately stable conditional on V . Then the global quantile will be appropriate across the shifted years. Intuitively, pixels with large member disagreements (hard pixels) would have larger scores. Dividing by $\hat{a}(V)$ will equalize the scores, so that hard and easy pixels are all scored equally, being precise on easy inputs and conservative on difficult ones. By using a difficulty signal (epistemic uncertainty) that is specifically “shift-aware,” we expect that the score becomes shift-aware as well.

Chapter 3

METHODOLOGY

3.1 Data

This study uses imagery of the estuary in Morro Bay, California. The data was collected between 2018 and 2022. Eelgrass was mapped using a DJI Phantom 4 Pro drone with a high-resolution camera. Data was collected between the months of November and January (of the following year) during days of extreme low tides, when eelgrass is exposed. Flight times were approximately 1 hour due to limitations of tide duration. High-resolution estuary-wide orthomosaics were acquired (~5000 images per year) and manually annotated to delineate between eelgrass and non-eelgrass polygons by a team of GIS specialists, as mentioned in Tallam et al. (2023). This includes iterative validation checks, resulting in ~1000 hours spent per year. The resulting polygons were rasterized to match imagery, resulting in pixel-wise labels for semantic segmentation.

For model development, imagery from 2018-2021 is treated as in-distribution (ID), while imagery from 2022 is treated as temporally out-of-distribution (OOD). For 2022, instead of fully labeling the orthomosaic (which, as mentioned, can be very expensive), a set of 1000 randomly sampled points was labeled as either eelgrass or non-eelgrass. The 2018-2021 data was tiled into chips of size 448x448 pixels and a stride of 224 pixels, resulting in 50% overlap between chips.



(a) Original image chip

(b) Corresponding eelgrass segmentation label

Figure 3.1: Example 448×448 px tile from the drone orthomosaic and corresponding manually digitized eelgrass label.

2021 was chosen as the *calibration year* as it is the most recent pre-2022 year. For 2021, the data was partitioned into four subsets: 1. a train subset (together with 2018-2020), 2. a small temperature-scaling subset to estimate the models’ over- or underconfidence in their softmax probabilities, 3. a CP calibration subset to compute conformal scores and quantiles, and 4. an evaluation subset to test in-distribution coverage. The OOD 2022 evaluation set is thus never used in training, temperature scaling, or conformal calibration.

3.2 Model Training and Inference Pipeline

3.2.1 Segmentation Ensemble

To obtain the epistemic uncertainty proxy and achieve high-accuracy predictions, we trained a heterogeneous deep ensemble of semantic segmentation models to increase model diversity. In total $K = 6$ models were added to the ensemble. Of the 6, 2 of each model architecture were chosen. These architectures include DeepLabv3+, UNet, and two variants of Segment Anything Model (SAM) that were fine-tuned with low-rank adapters (Sam-LoRA). Each

model type had one multi-year (2018-2022) model in the ensemble and a single-year (2021 only) model. Training was performed using dual NVIDIA RTX 3090 (24 GB VRAM) GPUs, and took approximately one week for 4-year models and three days for single-year models.

The models were trained using the ArcGIS Learn Python module. For the two DeepLabV3+ models, we used a ResNet-101 backbone pre-trained on ImageNet, whereas the U-Net models, by default, use a ResNet-34 encoder. SAM-LoRA uses a pre-trained SAM encoder (vision transformer-based model) and inserts low-rank adapter layers to adapt the model for eelgrass segmentation, without fully tuning the model.

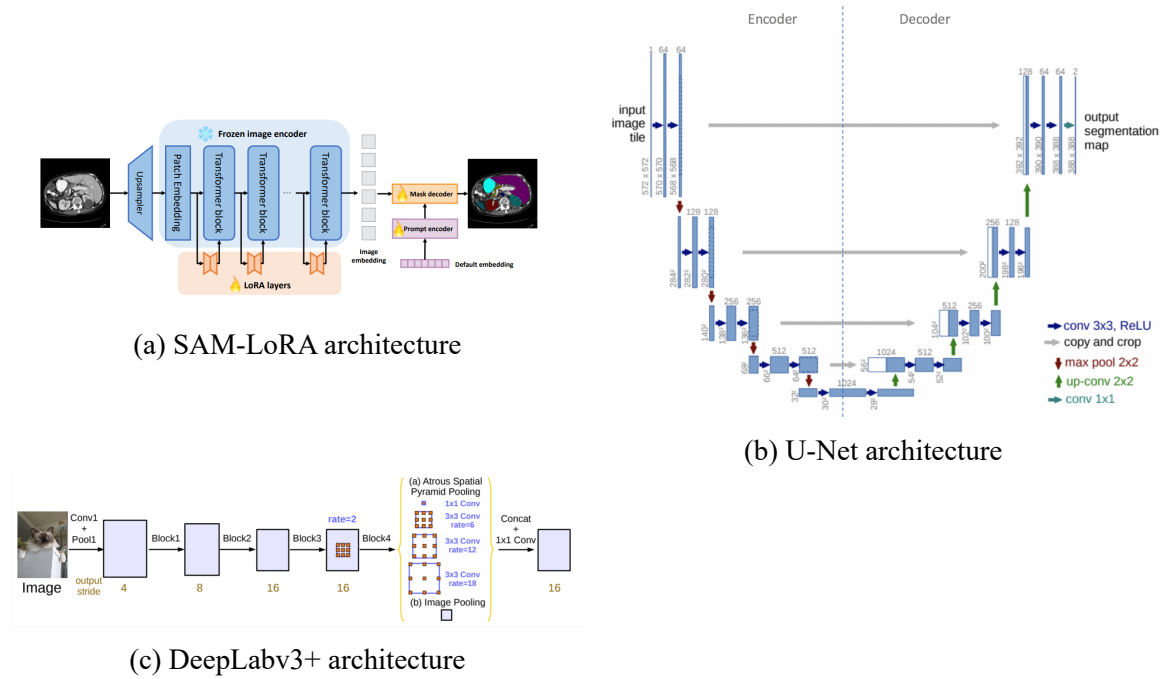


Figure 3.2: Model architectures included in the ensemble. Images from the ArcGIS Learn developer guides.

Models were trained for 20 epochs, and using a held-out validation set, the best checkpoint (best validation loss) was selected to be included in the ensemble.

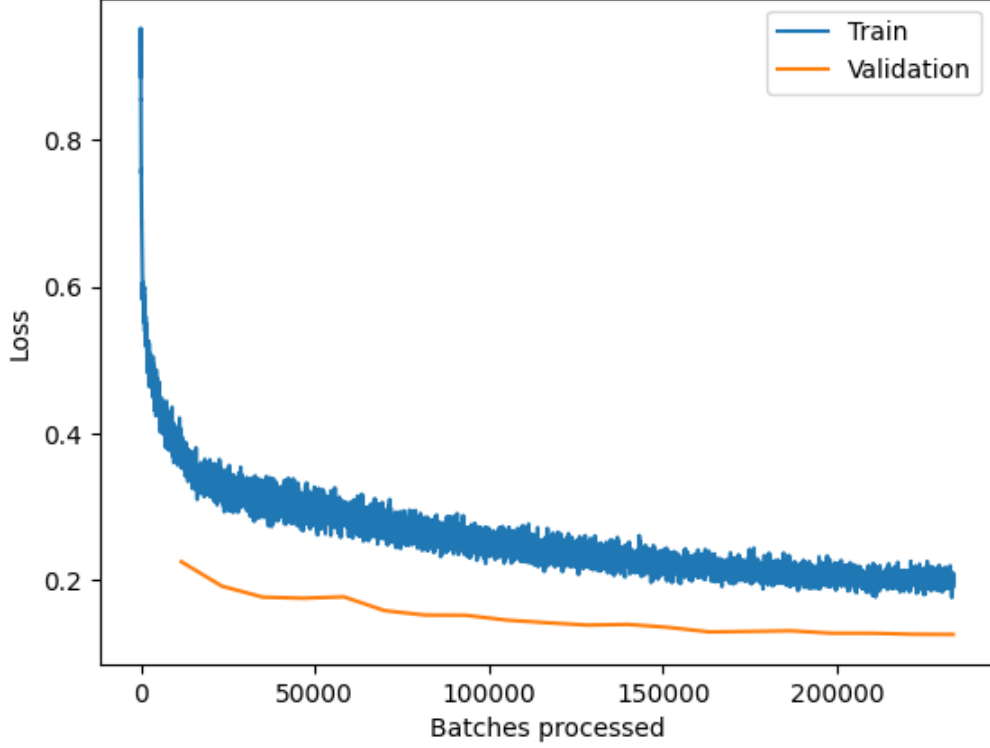


Figure 3.3: Example training and validation loss curve for DeepLabV3 4 year model

3.2.2 Probability Calibration and Aggregation

At inference time, each of the models produces per-pixel logits, which are then temperature-scaled, converted to class probabilities, and then aggregated. Temperature scaling calibrates the model confidence using a scalar temperature $T_k > 0$, which is calculated by minimizing the negative log-likelihood on the temperature-scaling calibration subset (Guo et al., 2017). For each model f_k , producing logits $z^{(k)}(x)$, probabilities are then converted using temperature scaled softmax

$$p^{(k)}(x; T_k) = \text{softmax}\left(\frac{z^{(k)}(x)}{T_k}\right).$$

We then compute the ensemble-averaged calibrated probability for class y at pixel x ,

$$\hat{p}_y(x) = \frac{1}{K} \sum_{k=1}^K p_y^{(k)}(x; T_k).$$

The ensemble prediction is thus the argmax aggregated probability. Each model’s individual probability output is also retained for computing the epistemic uncertainty proxy.

For each labeled point x , we compute the epistemic proxy based on member disagreement for the true class,

$$V(x) = \text{Var}_k(p_y^{(k)}(x; T_k)), \quad y = y_{\text{true}(x)}.$$

To assess how the proxy relates to errors, we use selective accuracy curves and correlation analyses. This was done with the variance scores (our epistemic proxy), as well as entropy and mutual information uncertainty scores for comparison. However, these were not chosen to be tested in conformalizing equations as they mix aleatoric and epistemic uncertainties and are implicitly used in the probability-based nonconformity scores $1 - \hat{p}_y$. For a given uncertainty score $u(x)$, labeled points are sorted by decreasing $u(x)$. We then iteratively recompute accuracy scores on smaller and smaller subsets of less uncertain points. For an informative uncertainty metric (that ranks errors well), we should observe a monotonic increase in accuracy as coverage decreases.

3.3 Split CP for Classification

The ensemble model is treated as a single probabilistic classifier for conformal prediction. We apply split conformal prediction to obtain per-pixel prediction sets with finite-sample coverage under exchangeability (Angelopoulos & Bates, 2022). For each labeled pixel x and class y , we define a nonconformity score such as the vanilla $S(x, y) = 1 - \hat{p}(x)$, which is large when the model assigns a low probability to the true class (nonconforming). On the 2021 calibration subset C of size n , we compute nonconformity scores on all pixels.

Then sort the pixels and compute the finite-sample corrected empirical $(1 - \alpha)$ quantile $\hat{q}_\alpha = \text{Quantile}_{\lceil (n+1)(1-\alpha) \rceil / n}(\{s_i\})$ for a target miscoverage level. For a new pixel, the CP set is

$$C_\alpha(x) = \{y \in \{\text{eelgrass}, \text{background}\} : S(x, y) \leq \hat{q}_\alpha\}.$$

Under the assumption that the calibration and test data are exchangeable, marginal coverage is guaranteed: $\Pr\{Y \in C_\alpha(X)\} \geq 1 - \alpha$. CP sets are computed for both a held-out 2021 ID set and the 1,000 labeled 2022 OOD points. We used the vanilla score to provide a baseline for OOD data coverage compared to the variance-aware methods.

3.4 Variance-Aware Score Normalization

3.4.1 Parametric linear shrink

3.4.2 Nonparametric normalization

3.5 CP Evaluation

Set Composition, Spatial Equality, and Conditional Coverage

Chapter 4

RESULTS

4.1 Ensemble Results

4.2 In-Distribution Evaluation

4.3 Temporal OOD (2022)

global coverage and set composition

4.4 Class Conditional Coverage

4.5 Spatial Robustness within 2022

4.6 Sensitivity

Chapter 5

DISCUSSION

Chapter 6

LIMITATIONS

Chapter 7

CONCLUSION

7.1 Computational Details

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APPENDICES

Appendix A

SUPER COOL FANCY FUNCTION